

The glycemic index: methodology and clinical implications^{1,2}

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ABSTRACT There is controversy regarding the clinical utility of classifying foods according to their glycemic responses by using the glycemic index (GI). Part of the controversy is due to methodologic variables that can markedly affect the interpretation of glycemic responses and the GI values obtained. Recent studies support the clinical utility of the GI. Within limits determined by the expected GI difference and by the day-to-day variation of glycemic responses, the GI predicts the ranking of the glycemic potential of different meals in individual subjects. In long-term trials, low-GI diets result in modest improvements in overall blood glucose control in patients with insulin-dependent and non-insulin-dependent diabetes. Of perhaps greater therapeutic importance is the ability of low-GI diets to reduce insulin secretion and lower blood lipid concentrations in patients with hypertriglyceridemia. *Am J Clin Nutr* 1991;54:846–54.

KEY WORDS Diabetes, diet, carbohydrate, glycemic index

Introduction

The systematic classification of foods according to their glycemic responses was first undertaken by Otto and Niklas (1), who, after testing individual foods, allowed their incorporation into the diabetic diet in amounts inversely proportional to their glycemic responses to keep the glycemic impact of the diet constant. The glycemic index (GI) was developed independently as a classification of the glycemic effects of foods to supplement information about chemical composition given in food tables (2). It was reasoned that knowledge of the glycemic effects of individual foods might be of use in understanding the physiologic effects of whole diets. Unexpected differences between the GI values of different foods helped to highlight the importance of food factors not given in food tables such as food form, particle size, the nature of the starch, food processing, and antinutrients, which may have large effects on the physiologic properties of foods.

It was proposed that the GI should fulfill four criteria to be of clinical utility: 1) consistency of values for the same food across space and time, 2) application in individual subjects, 3) application to mixed meals, and 4) demonstration of clinically significant therapeutic improvements by practical dietary changes (3, 4). Our purpose is to address these criteria in light of data that have appeared since they were proposed. In addition, the methods for calculating and applying the GI values of foods will be defined. The GI is not the only valid way to analyze blood glucose responses. However, it is recognized that methodologic factors may influence profoundly the interpretation of glycemic-

response data (5). Thus, valid conclusions regarding the clinical utility of classifying foods according to their glycemic responses can be obtained only with appropriate and consistent methods.

Glycemic index methodology

Portions of test foods and white bread containing 50 g available carbohydrate are fed to normal or diabetic subjects in random order on separate occasions after an overnight fast. To reduce variability the standard food (white bread) should be repeated at least three times by each subject (6). Capillary finger-prick blood samples are taken for normal subjects fasting and at 15, 30, 45, 60, 90, and 120 min after the start of the test meal, and for diabetic subjects fasting and at 30-min intervals for 3 h. The normal dose of insulin or oral hypoglycemic agent, if any, is taken after the fasting blood sample and 5–10 min before starting to eat the test meal. The area under the glycemic-response curve for each food is expressed as a percent of the mean response to the standard food taken by the same subject, and the resulting values are averaged to obtain the GI value for the food.

If an individual subject's GI value for a test food is an outlier, ie, > 2 SD from the mean, it could be due to an unrepresentative response to the standard or test food or to the subject having a true idiosyncratic response. An unrepresentative response to the standard food is suspected if the subject's GI values for other test foods are also outliers in the same direction, or if there is large variability for the repeated standard tests, with one very low or very high value. If the response to the standard food seems representative, repeated testing of the test food may be carried out. The outlying result can be discarded if two further tests result in a GI value similar to that for the rest of the group but must be suspected as real if a repeated test produces a similar result.

Important variables that affect the GI value obtained include food-portion size, choice of standard food, repeated testing of the standard food, frequency and length of time of blood sampling, and method of area calculation. Other factors include the method of blood sampling; subject characteristics, eg, age, sex, body fatness, glucose tolerance status; dose and timing of insulin

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or oral hypoglycemic agent; the degree of diabetes control, and, particularly in insulin-dependent diabetes mellitus (IDDM) subjects, the fasting blood glucose value on the day of the test. Although subject characteristics, treatment, and degree of control may have major effects on the absolute glycemic response obtained, if they are standardized they appear to influence the response to all foods similarly and so have only small effects on the resulting GI value, usually by influencing the variability of glycemic responses.

Food-portion size

Food-portion size has a major effect on the GI value because glycemic responses are related to the carbohydrate load (2, 7). The dose response for an intermediate GI food, bread, and for glucose appears to be nearly linear up to 50 g available carbohydrate, but the dose response flattens between 50 and 100 g. The GI is based on 50 g available-carbohydrate portions (2). Thus, food-portion sizes based on food tables or food analyses that do not give accurate dietary fiber values may result in portion sizes that contain < 50 g available carbohydrate, leading to an underestimate of the GI value. This is especially true for high-fiber foods such as legumes and fruit. Foods should be weighed dry because the water content of cooked foods may vary markedly. For certain foods, such as potatoes, food tables may give values for the cooked food only, making consistent portion sizes difficult to obtain. Differences in the weights of potatoes used by various investigators, ranging from 250 (1) to 317 g (8), may partially account for the variability of GI values obtained.

Choice of standard food

Originally, the GI was based on 50 g glucose as the standard (ie, GI_g , where the GI_g of glucose = 100) (2). Because of concerns that excessive sweetness and the osmotic effect of glucose solutions could lead to delayed gastric emptying, it was suggested that white bread be used as the standard (ie, GI_{wb} where the GI_{wb} of white bread = 100) (9). Another advantage of white bread over glucose is that bread stimulates more insulin relative to the blood glucose response than does glucose (10). Thus, when compared with a bread standard, the insulinemic index of different foods is related to their GI_{wb} values (11). However, the insulinemic index of foods based on a glucose standard are not related to their GI_g values (10). It has been suggested that white bread is not a good standard because it does not give the same GI value relative to glucose in all situations. This was especially so in one particular study where the GI_g of white French bread was 97 (10). Nevertheless, this may have been a specific effect of French bread because in nine other studies the mean GI_g of bread was 73 with an SD of only 5 (CV = 7%) (12).

GI_{wb} values are higher than GI_g values by a factor of $100/73 = 1.37$, where 100 is the GI_g value of glucose and 73 is the mean GI_g value of white bread from nine different studies (12). To ensure the validity of a proportional adjustment of GI values based on different standards, normal subjects tested 21 foods; repeated tests were conducted with both glucose and bread to allow GI_{wb} and GI_g values to be calculated. The mean GI_{wb} was virtually identical to the mean of the adjusted GI_g values, with a correlation coefficient of $r = 0.978$ (12). This finding suggests

that other standard foods could be used but to allow comparison with GI_g or GI_{wb} values, the standard chosen should be compared carefully with white bread or glucose before doing other tests.

Number of standard tests

To reduce variability it has been suggested that each subject should do at least three bread tests with the mean result being used to calculate the GI values of foods. The GI is the ratio of two independently variable values. As the variability of the values from which any ratio is calculated increases, the distribution of the ratio becomes skewed and the mean increases, even if the original values are normally distributed. Thus, when the mean of three white-bread response areas is used in GI calculations, the mean, variability, and skewness of the resulting GI distribution are reduced.

These effects can be illustrated by mathematical modeling. Two sets of 1000 normally distributed random numbers were generated by computer with a CV set at 15% [the mean CV of glycemic responses for subjects with non-insulin-dependent diabetes (NIDDM)] and means set at either 800 (representing the glycemic-response areas for a hypothetical food, F, with a GI of 80) or 1000 (representing the response areas for bread, B). The values of the two sets of numbers obtained were 796 ± 125 ($\bar{x} \pm SD$) and 993 ± 151 for F and B, respectively. GI values were calculated from the B and F values in order from first to last. The distribution of the resulting GI values was skewed with a mean of 81 and a CV of 22% (Fig 1). The skewness virtually disappeared and the mean and CV were reduced to 80% and 18%, respectively (Fig 1), when the mean of three consecutive B values was used to calculate the GI values.

These effects are relatively small when the CV of glycemic responses is 15%, as for subjects with NIDDM (6). However, they are more important when the CV of glycemic-response areas becomes 30%, as is the case for patients with IDDM (6). When the mathematic model described above was applied to subjects with IDDM by generating sets of numbers for B and F with a CV of 30%, the resulting mean GI value was reduced from 86 to 81 and the CV from 46% to 37% when the mean of three B values was used instead of only one (Fig 2).

Length of time of blood sampling

The effect of the length of time of blood sampling on the relative glycemic area of different foods was reviewed (5). Because the GI is calculated from the area under the glycemic-response curve, ignoring the area beneath the baseline, ideally, measurement need only be continued until the blood glucose response returns to baseline. Extending measurement any longer tends to reduce the differences in GI between foods, especially in normal subjects where foods with a high GI result in high peak rises of blood glucose followed by an undershoot of the baseline, whereas foods with a low GI have a low peak rise but tend to remain slightly above the baseline for a prolonged period of time. If measured for 4 h in normal subjects, these types of blood glucose curves may have the same area, despite markedly different effects on insulin, counter-regulatory responses, and insulin resistance (13). The latter three factors are related to the rate of absorption of carbohydrate, which in turn is related to

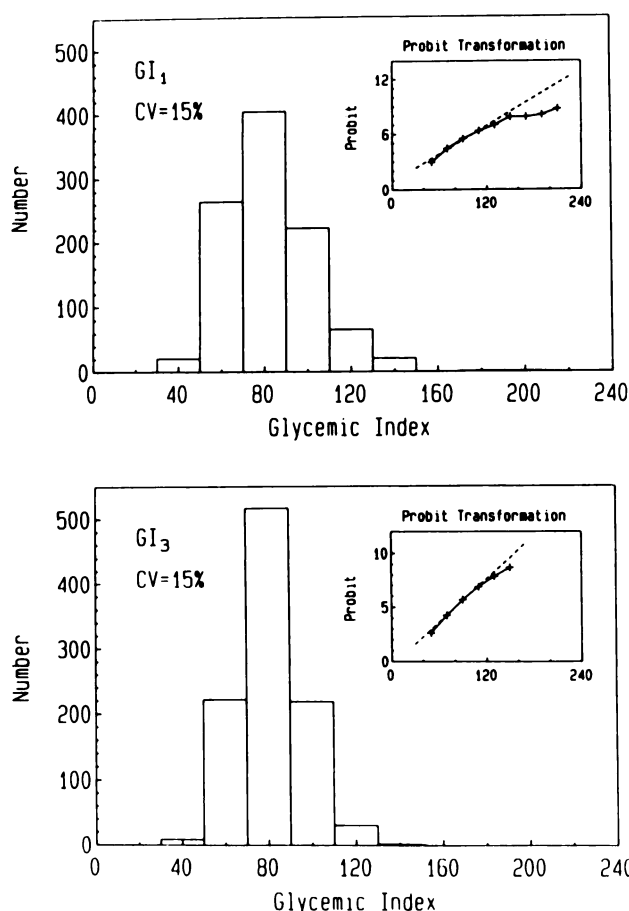


FIG 1. Mathematical model of glycemic index (GI) in NIDDM. Normally distributed random numbers with a CV of 15% were generated by computer, one set representing the glycemic-response areas to bread ($\bar{x} = 1000$) and the other the responses to a food ($\bar{x} = 800$). Distribution of 1000 GI values when one bread value (GI1, top) or the mean of three bread values (GI3, bottom) was used to calculate the GI. Inset panels: deviations of the solid lines (observed probits) from the dotted lines (expected probits) represent deviations from the normal distribution.

the incremental area under the relatively early part of the blood glucose-response curve ≤ 2 h in normal subjects. For diabetic subjects 3 h was chosen as a compromise between what is ideal to allow the blood glucose response to return to baseline (5 h) and what is practical for subjects testing many foods (5).

Calculation of area under the curve

Several methods have been used to calculate the area under the glycemic-response curve (14). Given the same blood glucose data, different methods may result in markedly different areas and GI values (Table 1). The GI is based on the area under the blood glucose-response curve above the baseline only (15). The formula is as follows:

$$\frac{At}{2} + At + \frac{(B - A)t}{2} + Bt + \frac{(C - B)t}{2} + Ct + \frac{(D - C)t}{2} + Dt + \frac{(E - D)t}{2} \dots \text{etc}$$

where A, B, C, D, and E represent positive blood glucose increments; t is the time interval between blood samples. If the blood glucose increment D is positive (ie, greater than baseline) and E is negative (ie, less than baseline) only the area between D and E above the baseline is included. If value E occurs t min after value D, a straight line drawn between points D and E crosses the baseline at time T after D, where $T < t$. The area above the curve between D and E is thus $DT/2$. Because $T/t = D/(D + |E|)$ (where $|E|$ = absolute value of E), therefore, $T = Dt/(D + |E|)$. Thus, $DT/2 = D^2t/2(D + |E|)$. The overall equation simplifies to

$$\text{Area} = (A + B + C + D/2)t + D^2t/2(D + |E|)$$

(The authors will be pleased to supply the algorithms required for computer or programmable calculator on request.)

Effects of fat and protein

Fat and protein influence glycemic responses by delaying upper gastrointestinal transit (16) and increasing insulin secretion

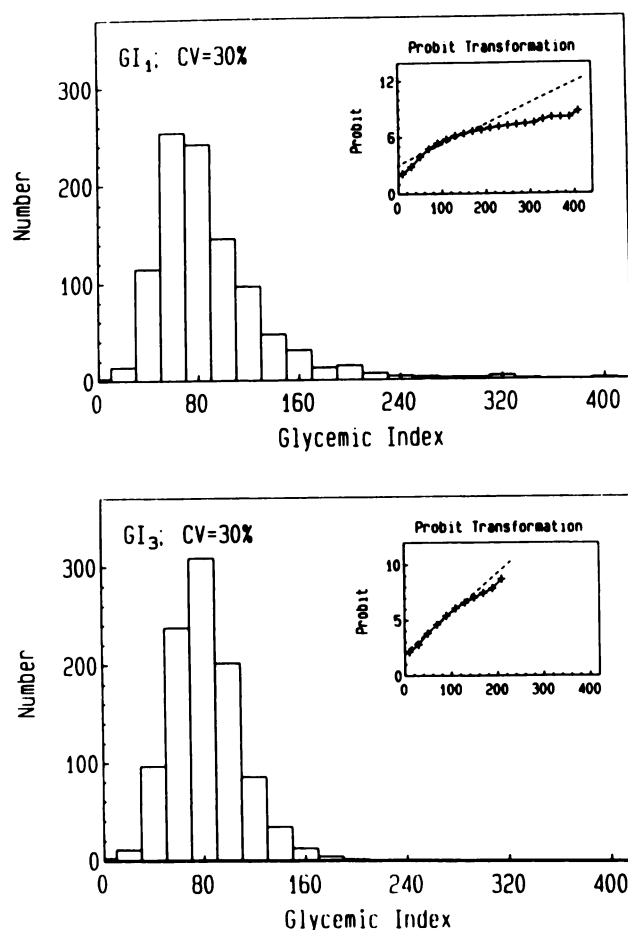


FIG 2. Mathematical model of GI in IDDM. Normally distributed random numbers with a CV of 30% were generated by computer, one set representing the glycemic-response areas to bread ($\bar{x} = 1000$) and the other the responses to a food ($\bar{x} = 800$). Distribution of 1000 GI values when one bread value (GI1, top panel) or the mean of three bread values (GI3, bottom panel) was used to calculate the GI. Inset panels: deviations of the solid lines (observed probits) from the dotted lines (expected probits) represent deviations from the normal distribution.

TABLE 1

Effect of different methods of calculating the area under the curve from the same blood glucose data on the GI in a diabetic subject being treated with insulin and in a nondiabetic subject

	Blood glucose									Area under the curve*		
	0 min	15 min	30 min	45 min	60 min	90 min	120 min	150 min	180 min	Inc	Net	Total
	<i>mmol/L</i>											
Diabetic subject												
Bread	10.1	—	13.7	—	15.0	14.0	12.8	11.1	8.9	475	465	2283
Lentils	9.8	—	12.0	—	12.7	11.2	10.2	9.0	8.1	203	158	1922
GI of lentils										43	34	84
Nondiabetic subject												
Bread	4.3	4.8	6.2	5.5	5.0	3.9	4.1	—	—	66	55	571
Lentils	4.3	4.8	4.8	4.3	4.4	4.5	4.7	—	—	29	29	545
GI of lentils										44	53	95

* Inc, incremental area under the glycemic-response curve, ignoring the area below fasting, as used for calculating the GI; net, incremental area under the glycemic-response curve subtracting the area beneath the fasting amount; total, total area under the glycemic-response curve.

(17), respectively. There is some evidence that the amounts of fat and protein required to have significant effects are large compared with the amounts normally eaten or advised in dietary recommendations (17, 18). Nevertheless, if the same amounts of the same fat and protein sources are added to each test food, there is no evidence that they affect the relative difference in glycemic response between the foods (10, 19). Recent data suggest that this may not hold if different types of fat and protein are added to the foods. There is some evidence that n-3 fats increase insulin secretion to a greater extent than do n-6 fats in normal subjects (20), although long-term consumption of n-3 fats increases blood glucose in diabetic subjects (21, 22). Protein increases insulin secretion because of the effects of amino acids. Thus proteins that are slowly digested, such as raw egg albumin, do not increase insulin secretion as much as rapidly digested proteins such as cottage cheese (23, 24). More work is required in this area. Nevertheless, different fats and proteins are not likely to influence markedly the utility of the GI under normal circumstances because the amounts of different proteins and fats consumed in normal meals are usually smaller than those required to obtain effects in experimental situations (25).

Blood sampling

The GI was based on measurement of glucose responses in whole capillary blood because this simple and relatively non-invasive method of blood sampling allows for extensive screening of foods. Glycemic responses in capillary blood are greater than those in venous blood or plasma and therefore may allow smaller differences in glycemic responses to different foods to be detected (26). For example, in normal subjects a mixed meal containing spaghetti had a capillary blood glucose response 37% less ($P < 0.05$) than one containing bread, but the difference in simultaneously obtained venous whole blood was only 19% and not significant (11).

Subject characteristics

Subject characteristics do not appear to have a major effect on the mean GI values of foods, except for a small increase (~ 5)

when comparing IDDM with NIDDM, a difference that is accounted for by the increased variability of glycemic responses in IDDM (27). Such a small difference is unlikely to be detected with < 60 subjects and does not affect the practical utility of the GI because the correlation coefficient of GI values for 21 foods in IDDM vs NIDDM was $r = 0.928$ ($P < 0.001$) (27). Characteristics that have been examined specifically and have no significant effect include normal vs diabetic subjects (9), NIDDM subjects on oral agents vs NIDDM subjects on insulin (28), children vs adults (29), rural African vs normal Western subjects (2, 30), and NIDDM subjects in good control vs NIDDM subjects in poor control (31).

Fasting blood glucose value

Paradoxically, the rise in blood glucose after a standard meal is inversely related to the fasting blood glucose value (FBG) in subjects with IDDM (32), but not in subjects with NIDDM (6). This effect increases the variability of glycemic responses in IDDM subjects and provides a rationale for stabilizing blood glucose values by using the artificial pancreas before testing foods (33, 34). However, screening many foods by using the artificial pancreas would be difficult because of the expense and subject time involved. In addition, even without the use of the artificial pancreas, the GI values of foods in IDDM and NIDDM subjects are similar (27).

Clinical utility of the glycemic index

Consistency of values across space and time

Eleven foods have been tested in at least three different centers using different groups of subjects (12); the mean $\pm$ SD and CV of the GI values are shown in Table 2. For four foods, most notably potato, the CVs are $> 20\%$. Much of the excess variation in potato is accounted for by reproducible differences in potato variety; russett potato used in four studies had a GI_{wb} of 116 ± 26 compared with 80 ± 13 for new potatoes ($P < 0.05$). The

TABLE 2
Glycemic index of foods tested in three or more centers*

Food	Mean $\pm$ SD	CV
		%
Whole-meal bread ($n = 8$)	100 $\pm$ 5	5
White spaghetti ($n = 5$)	67 $\pm$ 16	23
White rice ($n = 12$)	77 $\pm$ 12	16
Corn ($n = 6$)	80 $\pm$ 11	13
Corn flakes ($n = 4$)	121 $\pm$ 13	11
Oatmeal ($n = 5$)	89 $\pm$ 10	12
Potato ($n = 10$)	98 $\pm$ 26	26
Kidney beans ($n = 6$)	38 $\pm$ 9	23
Apple ($n = 3$)	52 $\pm$ 5	9
Banana ($n = 4$)	84 $\pm$ 14	17
Orange ($n = 3$)	59 $\pm$ 14	23

* The GI of white bread = 100. n = number of studies or centers; each study provided one value (12).

variability of GI values of foods tested in different parts of the world may be due in part to differences in testing methodology and to differences in food-portion size, processing, cooking, ripeness, storage, content of antinutrients, and nutrient-nutrient interactions (35). To some the field might appear too variable to allow meaningful interpretation. An alternative view is that much data is being acquired that will change our perception of food systems allowing predictions to be made on the basis of knowledge of the physiologic responses to foods. The surprising fact is that, despite all the unknowns, there is a broad measure

of agreement on the relative glycemic effect of many foods. The mean CV for the 11 foods tested in different centers is 16% (Table 2), similar to the variability seen in subjects with NIDDM doing repeated tests of the same food and less than that in normal subjects (CV = 22%) and subjects with IDDM (29%) (6).

Application in individual subjects

When a group of subjects tests a food once, the GI value obtained in each subject may vary considerably (4). However, this does not necessarily mean that there are real differences between the subjects because the variability could be the result of within-individual variation. To determine whether there are significant differences in GI values between subjects, each subject needs to test each food repeatedly. When a group of 12 heterogeneous diabetic patients took bread, rice, and spaghetti meals four times each, there was more than a fourfold range of absolute glycemic response areas between the different subjects (Fig 3) (36). However, when the results were expressed as the GI, there was no significant difference between the subjects (Fig 3). Thus, the variability in GI values in different subjects is largely due to within-individual variation. This implies that it is valid to apply the GI values for foods determined in one group of subjects to different individuals.

Application to mixed meals

The GI of meals containing several carbohydrate foods is expressed as the weighted mean of the GI values of each of the component foods, with the weighting based on the proportion of the total meal carbohydrate provided by each food (15). The approach can be extended to entire diets (37). A sample calculation is given in Table 3.

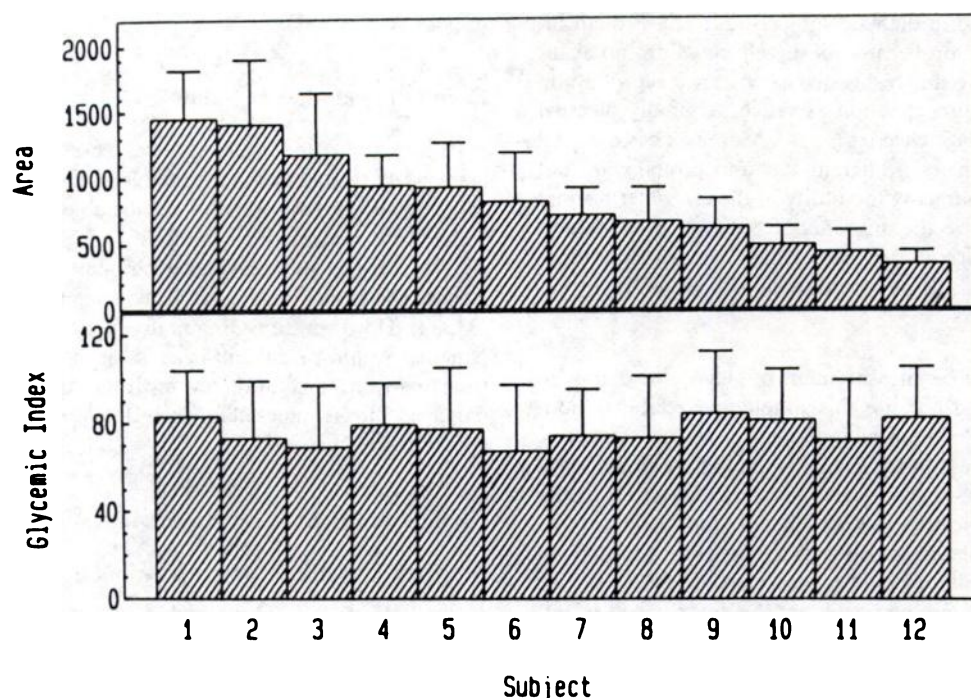


FIG 3. Glycemic-response areas (top) and glycemic-index values (bottom) of 12 subjects with diabetes after taking mixed meals containing either bread, rice, or spaghetti. Each meal was repeated four times; the GI was calculated by using the mean of the four bread response areas (36). $\bar{x} \pm$ SD.

TABLE 3
Sample calculation of mixed-meal GI*

	Carbohydrate	Food GI	Meal GI
<i>g</i>			
Meal 1			
All bran	30 (57.7)	73	42.1
Orange juice	16 (30.8)	59	18.2
2% Milk	6 (11.5)	48	5.5
Total	52 (100)	—	65.8
Meal 2			
Corn flakes	30 (57.7)	121	69.8
Orange juice	16 (30.8)	59	18.2
2% Milk	6 (11.5)	48	5.5
Total	52 (100)	—	93.5

* Data from 38. The GI of the All Bran meal is 70% that of the corn flakes meal; the observed glycemic response of the All Bran meal was $76 \pm 9\%$ that of the corn flakes meal ($P < 0.05$).

Values in parentheses represent the percent of total meal carbohydrate.

Quantitative prediction of relative glycemic responses

When individual foods have been pretested so that their GI values are known, the percent difference between meal GI closely predicts the percent difference between the mean meal glycemic responses (10, 39, 40). The group from Stanford, without pre-testing foods, concluded in each of three studies that the GI does not predict the relative difference between the glycemic responses of mixed meals (41–43). However, the relationship between meal GI and observed glycemic response from the Stanford studies is significantly different from that based on data from seven other groups in various parts of the world (34, 38–40, 44–48) where the regression line has a slope no different from 1 with a y intercept no different from 0 ($P < 0.001$; Fig 4).

Qualitative prediction: ranking glycemic responses

In individual subjects the GI can be used to predict which of two mixed meals of equivalent macronutrient composition will have the greater glycemic response. Assuming that glycemic responses are normally distributed, the chance of a correct prediction increases as the difference in GI between the meals increases and as the variability of glycemic responses from day-to-day in the subjects being considered decreases. Recently, the day-to-day variability of glycemic responses in NIDDM and IDDM subjects was determined and the resulting values were used to calculate the probability of correctly ranking glycemic responses for any given GI difference (Fig 5) (49). If the meals shown in Table 3 (GI difference = 28) were tested by an individual with NIDDM, there is a 91% chance that the All Bran (Kellogg Co, Battle Creek, MI) meal would have a smaller glycemic response than would the cornflakes meal (Fig 5). In a subject with IDDM, with more variable glycemic responses, the chance would fall to 82%. The difference in GI between the two meals such that there is a 95% chance of correctly predicting their glycemic response ranking is termed the predictive difference (PD). For subjects with NIDDM the PD is 34 whereas for subjects with IDDM, who have more variable glycemic responses, the PD is 50. If the subject repeats the meals n times or n subjects test each meal, then the PD decreases by a factor

of $1/(n)^{0.5}$. Thus, if four subjects with NIDDM test two meals with a GI difference of 16, there is a 95% chance that the meal with the lower GI will have a lower mean glycemic response. The PD values and probabilities given in Figure 5 were shown to be valid for groups of subjects (49) and for individuals (36).

Therapeutic effects of low-GI diets

Reducing the GI of the diet with no change in macronutrient or dietary fiber content was shown to result in modest but significant reductions in blood glucose concentrations in normal subjects (50) and in patients with IDDM (44, 51) and NIDDM (52), as measured by glycosylated albumin or hemoglobin. This was associated with reduced insulin secretion, assessed by urinary C-peptide excretion (50). Although low-GI diets do not have consistent effects on blood lipids in normolipidemic subjects, a modest reduction in diet GI of ~ 12 in patients with raised serum triglyceride concentrations reduces serum triglyceride by 20% and cholesterol by 9% (53). The GI may also have implications for nondiabetic and normolipidemic individuals because low-GI foods may induce higher satiety (28, 54, 55) and prolong endurance in athletes undertaking prolonged strenuous exercise (56). These studies suggest that the quality of dietary carbohydrate may influence the metabolic response to high-carbohydrate diets.

Mechanism of action

Several studies have shown that the rates of digestion of foods in vitro are related to their glycemic responses in vivo, suggesting

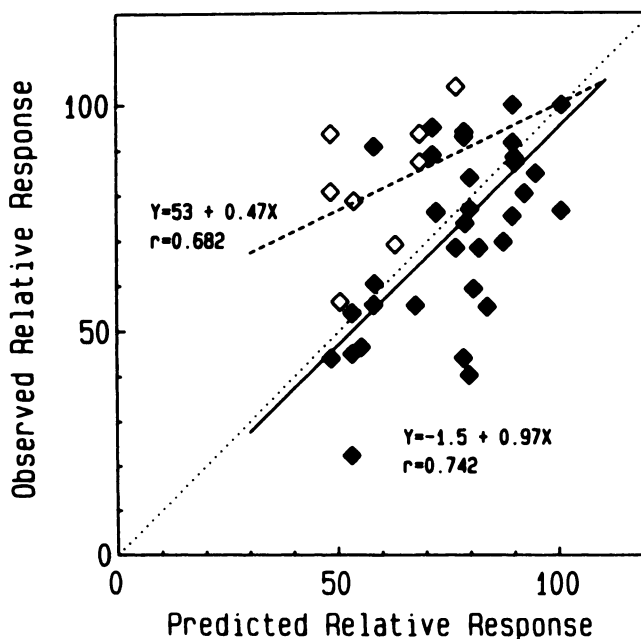


FIG 4. Relationship between predicted and observed glycemic responses for mixed test meals of similar composition taken by groups of normal or diabetic subjects. The predicted response was the meal GI for each meal expressed as a percent of the meal with the highest GI. The observed response was the incremental glycemic-response area expressed as a percent of the meal with the largest glycemic response. Line of identity (dotted line); data from 10 groups of test meals tested in seven different centers (■, —); data from three groups of test meals tested in Stanford (□, ---).

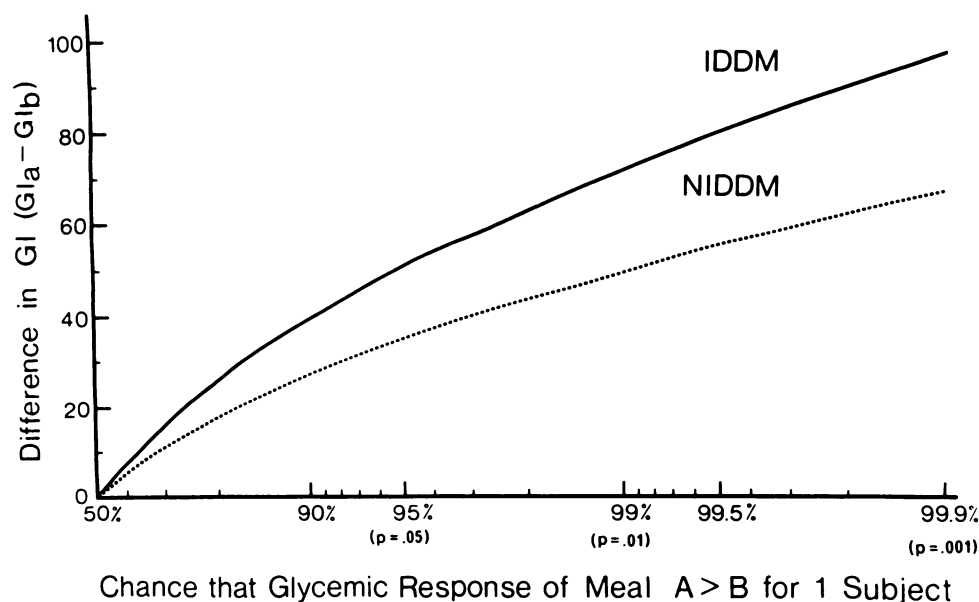


FIG 5. Probability that the glycemic-response area of meal A will be greater than that of meal B when each meal is tested once by an individual with insulin-dependent diabetes (IDDM) or non-insulin-dependent diabetes (NIDDM). The glycemic-index (GI) values of meals A and B are GIa and GIb, respectively. GIa and GIb are calculated from tables giving GI values of the constituent foods and GIa is $\geq$ GIb (49).

that differences in GI are due to differences in the rates of digestion and absorption of carbohydrate from different foods (57–59). Supporting this are data showing that glucose or food ingestion over a prolonged period of time, mimicking slow absorption, results in flat blood glucose and insulin responses (13, 60, 61). Reduced insulin concentrations may be the mechanism for the lipid-lowering effects of slowing absorption. When normal subjects consumed a metabolically controlled high-carbohydrate diet for 2 wk in 17 equal meals per day (nibbling), the mean serum cholesterol concentration was 8.5% lower than after eating exactly the same diet as three meals per day for 2 wk (61). The nibbling diet was associated with reductions of 20–28% in urinary C-peptide and day-long serum insulin concentrations; insulin is known to regulate the activity of 3-hydroxy-3-methylglutaryl coenzyme A (HMGCoA) reductase, the regulatory enzyme for cholesterol synthesis (62).

Conclusions

Recent studies support the clinical utility of the GI. Within limits determined by the expected GI difference and by the day-to-day variation of glycemic responses, and ranking the glycemic potential of different meals is possible. Although reducing the fluctuations in blood glucose after meals has only a modest effect on overall blood glucose control, it may be beneficial in some patients with diabetes. Of perhaps greater therapeutic importance is the ability of low-GI diets to induce useful reductions in blood lipids in hypertriglyceridemic patients.

High-carbohydrate diets are recommended for individuals with diabetes and hyperlipidemia but the type of carbohydrate is likely to be important in determining the metabolic response to such diets. Increasing carbohydrate intake with high-GI foods may increase blood glucose, insulin, and triglyceride concentra-

tions (63). However, increasing carbohydrate intake with low-GI starchy foods may allow carbohydrate intake to be increased without these unwanted effects.

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